

The Soulful Science

WHAT ECONOMISTS REALLY DO AND WHY IT MATTERS

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What Makes Economies Grow?

Economic growth is an unusual phenomenon—at least, growth as we normally think about it now, “modern” growth with output expanding fast enough for one generation to have living standards far beyond those of the previous generation, and with extraordinary innovations which improve our health and longevity. It has, after all, only a 200-year record even in the West. What’s more, about twenty poor countries have been experiencing declining output per capita, negative growth, since the 1970s. Dozens of countries, including Nigeria and Pakistan, for example, have not yet attained the level of income per capita the United States had achieved by 1870. So one answer to the question posed in the title of this chapter must be that we’re not entirely sure, or every economy would be growing.

It might seem dubious to claim any theoretical certainty about the causes, the more time passes without all countries being able to achieve modern growth. Nevertheless, economics now has a theoretical analysis of growth which explains the empirical record reasonably well and also explains why it is difficult for an economy to get to the point where growth begins. It also exemplifies a characteristic new way of thinking about other issues in economics, such as international trade, industrial organization, or geographical location. This new generation of models has moved the economics profession decisively on from the familiar neoclassical framework which dominated for the second half of the twentieth century, a framework much criticized by noneconomists. The new approach in this particular context is broadly known as endogenous growth theory, and will occupy much of this chapter. These growth models are one example of this new generation of modeling in the fields of industrial organization, trade theory, and economic geography.

Readers can be forgiven a natural skepticism about the claim that, after a couple of centuries, economists have finally cracked the mystery of growth, the holy grail of policy makers. If so, why isn’t the whole world growing at last? The answer is that getting an economy to expand is more

like dancing than cooking. The frequently used metaphor of finding the right “policy recipe” is misleading. Theoretical knowledge is only a small first step toward practical success. It is not possible to set down a simple list of ingredients and a technique anyone could follow. Rather, getting an economy expanding in the way the rich countries already have for the past 200 years depends on a complex sequence of decisions and policies, involving many partners and depending on past choices, current resources, and pure luck.

As for why it took so long for economists to get close to a good theory of growth, one important reason has been the lack of data until quite recently. Over time, it has steadily become more apparent what the empirical predictions of a good theory need to be. The model of growth most economists used through the 1960s into the 1980s seemed to fit the then-available evidence reasonably well. Before the mid 1990s—as we saw in the last chapter—the extensive historical data and cross-country evidence were not available to test and inform the theory. Nor was there enough computer power to make testing theories easy. One of the themes throughout this book will be the importance of new data sets, cheap computer power, and the development of econometric techniques in the current renaissance in economics. That’s as true in growth theory as it is in other fields.

A second explanation for the renaissance in growth theory in the late 1980s lies in the renewed interest in the impact of new technologies when it started to become apparent that computer power was becoming cheap and pervasive. Unsurprisingly, the question of the role of innovation came to the fore once again. Growth in the rich industrialized economies had slowed down in the mid 1970s; might the new technologies improve the trend again? At the same time, politics in some big economies—step forward and take a bow Mr. Reagan and Mrs. Thatcher—had taken a decisive turn toward deregulation. Globalization was well underway, with rapid increases in trade and foreign investment. Meanwhile the planned, communist economies were clearly in crisis. The sheer pressure of events always shapes theory, and the unfolding of the 1980s led economists to think once again about the fundamental question of their subject: how does the process of growth work, and can it be influenced by policies? But let’s start well before the 1980s.

CLASSICAL GROWTH

With hindsight, the Industrial Revolution is clearly where the explanation for growth must be sought, but that wasn’t so obvious in the 1780s.

Needless to say, the earliest explanations about why output in Britain, the Netherlands, and France had started to grow much faster than population, and why real wages and living standards were growing, did not have much evidence to go on. Just about the earliest systematic theory was devised by French intellectuals known as the physiocrats or *economistes*, of whom the best known is François Quesnay. Quesnay seems to have been a remarkable character, born into a poor family, illiterate until the age of eleven, orphaned at thirteen. He taught himself medicine and eventually became, by way of being a country surgeon, personal physician to Madame de Pompadour, the influential mistress of Louis XV. He ended his days living at the Palace of Versailles, a member of the Académie des Sciences and a star among intellectuals, the Bernard-Henri Lévy of his day. Quesnay is remembered for his *Tableau économique*, a theoretical analysis of the circulation of the country's output among the different classes in society. The physiocrats believed there was scope for productive labor to create an agricultural surplus over and above the nation's subsistence needs. This surplus they attributed to the bounty of nature rather than to productivity gains achieved by the inventiveness of people. They concluded, wrongly with hindsight, that a large agricultural sector was therefore the foundation of economic growth. As we saw in the last chapter, a rapid move away from agriculture as it becomes more productive characterizes the transition to economic growth. Productivity growth in one sector of the economy means it will shrink, rather than grow, as a proportion of the total. Fewer people can produce a given amount of output, so its employment share will shrink. If demand for the output rises by proportionately less than income (the elasticity of demand is less than 1), which is certainly true of food, then its share of total output will also shrink.¹

Empiricist British thinkers disagreed with the abstract French, despite the deep similarities in their philosophies, firmly rooted in the Enlightenment in both cases. One of the most influential was Robert Thomas Malthus. He was a country parson and Cambridge academic (although he had to resign his fellowship at Jesus College when he got married in 1804, going on to father three children, a modest number for the time). He first published his "Essay on the principle of population" in 1798—with hindsight, again, at exactly the moment in history when it ceased

¹ The same arguments apply to the tenacious superstition that manufacturing is now shrinking "too much" in countries like the United States and the United Kingdom.

to apply universally.² His logic was that if people were not starving, the "passion between the sexes" would lead to an increase in the rate of population growth. More mouths to feed would at once return people to the serious business of scrambling for their next meal. Given that yields on land declined the more intensively it was farmed, or the more of the less fertile land was brought into use, at some point increased population growth would exceed growth in food supplies, and population growth would decline again. Not only logical, but also consistent with all of the past. It also earned economics the description "the dismal science" from historian Thomas Carlyle.

Malthus was one of a number of stellar British economists, including David Hume, David Ricardo, and of course Adam Smith, who laid the foundations for classical economics. Smith's views are now claimed by many others in support of their own. To most noneconomists, he is associated with free market, laissez faire, right-wing and minimal-state opinions.³ It is certainly true that, in common with many Enlightenment thinkers, Smith's aim was to justify individual freedoms and restrain arbitrary acts by the state. In terms of his contribution to understanding "the nature and causes of the wealth of nations," his central point is that growth depends on the division of labor or specialization. The scope for this will vary from country to country, depending on technology and organization as well as the government, although much modern comment on *The Wealth of Nations* focuses on this latter question of whether free competition or government regulation of industry has the upper hand. Smith noted that industry in general offered more scope than agriculture for specialization, and that the richer countries had more industry than the poorer ones. However, even though Smith knew James Watt, who spectacularly improved the steam engine, *The Wealth of Nations* largely ignores the technological innovations of the time—such as the flying shuttle, spinning jenny, and mule—and is rather scathing about innovators in general. Perhaps this is not surprising: there were only twenty to thirty factories big enough to employ as many as 300 workers when the book was published in 1776. Instead, Smith saw growth resulting from the process of exchange in the context of a market, organized by that famous "invisible hand." The fruits of specialization in labor could

² And indeed, Malthus realized this and changed later editions.

³ See Peter Dougherty's *Who's Afraid of Adam Smith: How the Market Got Its Soul* for an outstanding analysis of what's wrong with this caricature.

be amplified by exchanging with others in trade. He was right, but he overlooked the role of technology in the tale.

Adam Smith seems to have been, like so many economists today, somewhat awkward socially, happiest poring over his books (there being no computers available at the time). Who knows what he would have made of the modern cottage industry selling ties and mugs bearing his portrait, raising funds for the cause of *laissez faire*. Yet his work had an immediate impact on opinion and policy at the time, and he remains one of the most famous economists ever. His analysis was greatly extended and updated by the less-famous David Ricardo and John Stuart Mill. Ricardo made familiar the notion of diminishing returns: that bringing additional acres of land into use would generate a diminishing increase in output as a result. Ricardo also extended the analysis of the benefits of exchange to the context of international trade, introducing here the idea of comparative advantage.⁴ (The next chapter will return to questions of trade and openness.)

Classical economics was summed up in the bible of late-nineteenth-century economics, John Stuart Mill's *Principles of Political Economy*. It was published in 1848 and was still in use as a textbook in 1900. The analysis of growth was not one of its strengths either. Mill wrote disdainfully: "I am not charmed with the idea of life held out by those who think that the normal state of human beings is that of struggling to get on." He argued that the government should limit the pace of technical change, and was in favor of cooperatives and profit sharing. Like anyone with a shred of sensitivity and imagination, he was clearly moved by the horrors of rapid industrialization, the poverty and misery so visible in the thrusting new cities of Victorian Britain.

So too were Friedrich Engels and Karl Marx. It stirred them to write the *Communist Manifesto*. Engels was a wealthy cotton mill owner, so he knew at first hand observing his own workforce what others discovered from novelists like Charles Dickens and Elizabeth Gaskell. Marx of course drew rather more radical policy conclusions from his observations of the process of industrialization. But in contrast to Mill, and the mainstream economics tradition, he did focus on the process of growth, giving a central role to the accumulation of capital (hence the title of his weighty text, *Das Kapital*), and specifically to technology. This makes him seem rather modern. Paradoxically, there was a revival of interest in Marxist

⁴Although the passages in his work on comparative advantage were almost certainly written by James Mill (see Blaug 1996, p. 132).

economics in the dot-com 1990s, just after the collapse of communism, precisely because of this emphasis on technological innovation. Indeed, the relevant chapters on innovation in *Capital* refer to innovations in steam, iron, and steel, and use data provided by Engels on productivity gains in a real cotton spinning mill. Marx as a classical economist was rather prescient, whatever one's views about him as a political theorist.

This canter through the classics has brought the story to the early twentieth century. While there were disagreements between economists before then, it was at this point that the subject divided decisively into a mainstream and a minority view. One of the best-known economists to think seriously about the dynamics of capitalist economies was Joseph Schumpeter, who famously described the process of "creative destruction" whereby a whirlwind of technological innovation sweeps away old methods and businesses, introducing more productive new ones; but his was a minority interest. (I return to Schumpeter in chapter 7.) The mainstream became increasingly formalized and mathematical.⁵ It rested the analysis of economic behavior on marginal utility, and the utility- or profit-maximizing behavior of individual economic "agents." So too, in a formal, mathematical way, did the postwar "neoclassical synthesis" set out by Paul Samuelson in his *Foundations of Economic Analysis* in 1947. Yet this mainstream was essentially a static analysis because it was focused on the concept of equilibrium, the state in which the economic system is at rest. For a century, the mainstream neglected innovation and economic growth, only returning to the serious analysis of growth twenty years ago.

NEOCLASSICAL GROWTH

Let's turn now to a more recent era, the early 1950s. A computer was the size of a football stadium, and the total computer processing power available in the world amounted to the power of a single chip in any consumer electronic item today. National accounts were newfangled and available for just a few countries. Historical data still lay in scattered library stacks. How could an economist begin to think about the process of growth? Since the end of World War II, growth rates in rich countries have varied from year to year, and different countries have experienced different average growth rates. The mid 1970s brought a watershed, with

⁵One controversial and certainly rather tendentious explanation is that late-nineteenth-century economists were trying to make economics just like physics (see Mirowski 2002).

a faster average growth rate before than afterwards. However, the range of variation between countries and within countries over time is vastly greater for the poorer economies. To mangle Tolstoy's famous phrase, all rich economies are alike; it is poor ones that differ. To an economist in a European or American university looking at experience close to home in the postwar era, it was reasonable to approximate the growth performance of the rich economies as 2-4% a year for most countries and most years, a pattern that has not changed in fifty years. Of course, there has been variability over the course of the business cycle—booms and recessions—so a theory of the ups and downs has been needed on top of the theory of long-run growth. But there was a good workhorse theory of economic growth to explain the long-run stability, initially developed by Nobel laureate Robert Solow, explored and extended by Edward Denison, David Cass, and Tjalling Koopmans in the following years (Solow 1956). Its essence was, reasonably enough, that the growth of outputs depends on the growth of inputs, and on how effectively these inputs are used. This neoclassical model, newly systematized in the postwar years, was the staple until the renewal of interest in growth theory in the late 1980s.

One of its key principles was the concept of diminishing marginal returns. In consumption, this meant each extra apple consumed brought less utility to its eater than the previous apple. In production, it meant each extra unit of input used—land, labor, or physical capital—produced a smaller increment in output. The best land is used first so extra fields have lower crop yields; workers get tired; machines which are run for longer will break down.

Clearly, more inputs are needed for an economy to produce more output. This is bound to be the starting point for any explanation of growth, the accumulation and use of physical capital and human labor. Accumulating physical capital requires that the economy consumes less than today's total output in order to save and invest in new machinery. Human labor depends obviously on the number of workers, how hard they work, and also on their capabilities (which become more important the more jobs involve skills other than muscle power). Decisions to accumulate physical or human capital will depend on the incentives to save rather than spend, work rather than play. Given these decisions, the amount of output produced will increase as the inputs increase, but in diminishing increments. The relationship between inputs and outputs is known as the production function, and will depend on the state of technology or know-how. Some techniques will get more output than others from a given amount of each input (known in the jargon as

factors of production); and every technology will require a certain ratio between the different inputs. Say three laborers and three machines can produce twelve chairs a day. But diminishing returns to each input mean that if you add a fourth machine you get only fifteen chairs. Add an extra laborer instead and you get fourteen. But add a machine *and* a laborer, and you'll get sixteen chairs.

What will the aggregate growth rate be for the economy? The answer depends on how fast the workforce and the stock of machines can grow. So how fast is population growth? How much is the economy as a whole saving and investing, and how rapidly does existing capital equipment depreciate and need replacing to keep the capital stock intact? In addition, at what pace does technology progress, as innovations (a better irrigation system, a more efficient dynamo) permit the same amount of inputs to deliver more output?

To start with, let's concentrate on the accumulation of capital. The accumulation of capital seems pretty fundamental to capitalism, and it is natural to think that the rate of saving and investment in new equipment will be decisive for the economy's performance. So assume that the rate of population growth and the pace of technical progress are given and constant. Now, as a thought experiment, suppose that saving and therefore investment in the economy increase significantly (we are assuming away the rest of the world, too, so all investment has to be financed by domestic saving). Extra capital is accumulated, and extra output produced, so growth has accelerated. But in the long run, unless the workforce is also growing faster, the faster growth will not be sustained: diminishing returns to capital will kick in and steadily reduce the gains in output due to each extra increment in capital. The *level of output* per person will be permanently higher than it would have been without the high savings and capital accumulation, but ultimately the *growth rate* will not be any higher. If the ratio of capital to labor is "too high," the return to investment will decline (in line with its diminishing marginal productivity), and capital accumulation will slow down. In the long term—in equilibrium, the economist's metaphorical concept of the state in which the economy will settle down after any disturbance—the growth rate of output will keep the ratio of inputs constant. The long-run growth rate will depend on those things we assumed as givens, namely population and technological change.

Needless to say, I've given a very simple description of the basic growth model. Even so, there is obviously something very unsatisfactory about the Solow model. In effect it explains how economies grow by assuming

that they grow. Growth of the labor force is taken as a given. Technical progress is a gift from heaven: the production function is also given. Both are *exogenous*. What's more, every economist handing down this model to the next generation pointed out this flaw. Nor did anybody pretend it was a rounded explanation of the historical experience of growth, despite its tractable equations. There was no room for the drama of the Industrial Revolution or the violent history of empire in what Solow described as "the land of the margin." And to be fair to him, his 1956 article was offering a clear improvement over the previous leading model of growth.⁶

Let's set the theoretical shortcomings to one side, however.⁷ How does the neoclassical model fare when it meets the evidence? Obviously, it ignores the overwhelming importance of innovation in the experience of economic growth. Yet the theory had in mind an economy just like that of the United States or the United Kingdom or Germany, a rich economy with a long growth record. In the 1950s and 1960s, although the rich economies were dissimilar, with some experiencing a postwar reconstruction surge in growth, it did not seem unrealistic to suppose that small differences in population growth or the available technology could account for underlying differences in output growth. Empirical research very soon established, though, that the neoclassical framework of the accumulation of capital, subject to diminishing marginal returns, could only explain a small proportion of actual growth in any country. When the inputs were added up, in almost every case their growth proved to be substantially lower than output growth, which therefore had to be explained as the result of "technical progress." In 1956, Solow attributed a full four-fifths of postwar U.S. growth in business output to "technical progress." Of course, this sum depends on how inputs are measured; but even adjusting for improvements in the quality of the inputs—and importantly in the quality of labor, such as more years spent at school on average—all studies have found that (much) more than half of output growth in the developed economies has to be attributed to technical

⁶This was the Harrod-Domar model, in which the economy could sustain growth only if the natural rate of growth, which depended on population growth, happened by chance to be the same as the "warranted" rate, which depended on how much people decided to save and invest. The marginal neoclassical framework introduced a mechanism for adjustment of one to the other.

⁷After all, one of the classic jokes in economics is, "That might be fine in practice, but it'll never work in theory."

progress, which is unexplained by the model (Solow 1956; Jorgenson and Griliches 1967; Jorgenson and Yip 2000).

It is worth explaining how the empirical studies do these calculations, which are known as *growth accounting*. Logically, output growth can be exhaustively divided into growth in inputs, plus a residual. The part of growth attributed to labor is its growth rate multiplied by its share in total output, and likewise for capital. The contribution of labor to growth will (or should) incorporate quality changes, such as better schooling, and productivity improvements specific to labor, such as more effective teamwork methods; and likewise for capital. The part left unaccounted for by growth and improvement in inputs is known as *total factor productivity*. It is usually identified with technical progress, although it will also include anything left out of the measurement of inputs, including quality changes not properly accounted for. Such omissions would lead to overestimates of the importance of technical change. On the other hand, productivity improvements due to innovation will increase investment in capital, so part of the growth attributed to capital will in fact be caused by technical change.

While the logic is straightforward, in practice it can be difficult to attribute all changes to the right categories. Even so, there are plentiful studies of levels and growth rates of total factor productivity in many countries. All find that (a) it accounts for a large part of output growth, (b) its level and growth rate vary significantly between countries, and (c) its growth varies over time (see OECD 2001; Helpman 2004).

What is more, as the historical data have become more readily available, it has become apparent that the rate of growth in the industrialized economies has accelerated as the centuries have passed. This can be seen from table 1.1. Although average growth from the 1970s to 1990s was slower than in the preceding decades, it was nevertheless much higher than growth in any period before World War II. Paul Romer (1986) calculated that the probability that growth in a later decade would be higher than growth in an earlier decade was at least 58% and up to 81% for each of the eleven industrial countries he investigated. Given that rates of accumulation have diminished over time, the neoclassical model would imply slowing rather than accelerating rates of output growth, unless the pace of technological progress has been accelerating.

So the evidence from within the rich economies points firmly to the inadequacy of the accumulation of inputs in the context of diminishing marginal returns as an explanation for growth. Theory needs to say something about the forces that shape technology and innovation.

Table 2.1. Average annual compound growth rate of GDP per capita (%).

	1913-1950	1950-1973	1973-2001
Western Europe	0.76	4.05	1.88
Western offshoots	1.56	2.45	1.84
Eastern Europe	0.89	3.81	0.68
Former USSR	1.76	3.35	-0.96
Latin America	1.43	2.58	0.91
Japan	0.88	8.06	2.14
Other Asia	-0.10	2.91	3.55
Africa	0.92	2.00	0.19

Sources: Maddison (2003). Western offshoots are the United States, Canada, Australia, New Zealand.

A second empirical problem for the Solow model is its inability to explain the differences between countries, especially when other, poorer countries are considered. The model implies that, if they have similar technology and population growth, an economy with a high ratio of capital to labor will grow more slowly than one with a low ratio. High capital-to-labor economies are the rich ones: past savings have raised the level of output per capita. So in other words, the theory would lead you to expect poor countries to grow more rapidly than rich ones, and to expect levels of income per capita to converge upwards over time. The reason is that as capital is scarce relative to labor in poor countries, the return to capital is higher, so there is an incentive to save more and accumulate more capital. There's scope for poor economies to catch up until they too attain the steady long-run growth rate.

This implication would have seemed reasonable looking at the postwar experience of the Western economies, with countries such as France and Germany indeed growing rapidly at first and gaining on the United States in living standards, then growing more slowly. However, it is not a good description of a larger group of countries over a longer period. As Robert Lucas has pointed out, if the difference in incomes per capita between the United States and India were due mainly to the difference in their stocks of capital relative to labor, the return to capital in India should be so high that investors would pile all their money into India and none into the United States. Of course, this does not happen (Lucas 1988). Nor has there been any sign of increased convergence since the 1980s, when international barriers to capital mobility started to vanish.

Not only has there been no sign of systematic catch-up by poor countries, but as table 2.1 and figure 2.1 show, the wide variation in growth

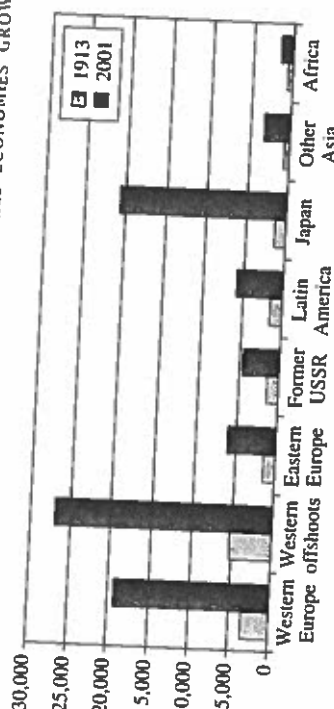


Figure 2.1. Level of GDP per capita (1990 international \$). Source: Maddison (2003).

rates has been getting wider still, and the divergence in incomes per capita has increased. Relatively few countries, including some western European countries in the postwar years and some Asian countries such as Japan and South Korea somewhat later, have followed the implied Solow pattern, growing very rapidly during a period of catch-up and attaining almost the same income per capita as the world leaders.

So in the data on growth and levels of output per capita, there is no evidence of convergence in living standards. There is almost no correlation between a country's initial income and its subsequent growth.

On the other hand, there is firm evidence of what economists call *conditional* convergence in incomes per capita. While there is no correlation between income per capita and subsequent growth, if income per capita is adjusted for ("conditioned on") differences between countries which affect its equilibrium level, then it is strongly negatively correlated with subsequent growth (Barro 1991; Barro and Sala-i-Martin 1995). For example, a poor country which grows very slowly is not converging on the rich countries; but its growth rate may be low because its people have little education, or its fertility and mortality rates are high, or the government makes unhelpful interventions such as spending too much on wars. Adjusting for the effect of such variables, a country's growth rate does tend to be systematically higher, the lower its starting point. By far the most important adjustment to make is the initial level of human capital, the number of workers and their quality.

All this suggests that considering the accumulation of capital is a good starting point for understanding growth. But an improved theory needs to do better at explaining the variations between countries, which the

evidence indicates will depend above all on their human capital; and it needs to explain what determines the rate of technical progress. The two *exogenous* variables in the Solow model, growth of labor input and technological change, need to be made *endogenous*.

POST-NEOCLASSICAL ENDOGENOUS GROWTH

May 2, 1997 dawned a brilliantly sunny spring day. Britain had just elected its first Labour government for eighteen years, and the Labour Party's celebrations were still continuing at the Royal Festival Hall by the River Thames in London. A short distance away in Number 10 Downing Street, Tony Blair was forming the first government of his eight-year-plus premiership. Gordon Brown was his Chancellor of the Exchequer (as Britain's finance minister has been known since 1170). Brown turned out to be—as even his political enemies acknowledge—one of Britain's most imposing chancellors. And one of his many gifts to the British people has been to introduce them to the term “post-neoclassical endogenous growth theory.”⁸ A theory must surely be good if it makes its way into political speeches. So what is it? The “post-neoclassical” bit means it is of 1980s vintage rather than 1950s. And “endogenous” means it offers an explanation for what had merely been assumed in the Solow model. In fact, there are two main flavors of the modern growth model, depending on which previously exogenous variable they focus on endogenizing. One important strand concentrates on human capital, the other on technological innovation. They originate, respectively, in papers by Robert Lucas (1988) and Paul Romer (1986, 1990).⁹

⁸In a 1997 speech, written for him by aide Ed Balls, he said: “You wouldn’t expect me to talk to you about post-neoclassical endogenous growth theory, and I won’t....” Conservative politician Michael Heseltine extended the joke with his pun: “It’s not Brown’s, it’s Balls’.” Economist Robert Barro had some years earlier been asked by *The Financial Times* (where Ed Balls worked at the time) to explain post-neoclassical endogenous growth theory to the British public, and was very miffed when the Conservative chancellor Kenneth Clarke poked fun at his article because it used the phrase. Reminiscing about the affair, Barro concluded: “I suppose that I will never understand British humor” (see Barro 1996, p. 22). However, the chasm in understanding may lie not so much between Brits and Americans, or even Conservative and Labour politicians, as between economists and others.

⁹See Warsh (2006) for a history of endogenous growth theory, citing these papers and many other relevant references. Warsh’s narrative centers on Paul Romer and consequently downplays the contributions of Lucas and economists working on similar models in other areas of economics, such as Avinash Dixit, Joseph Stiglitz, and Paul Krugman.

The antecedents of these new growth models lay in research in other branches of economics—industrial organization and trade theory—in the late 1970s and early 1980s. The new techniques they introduced abandoned the simplifying assumptions made in all earlier work that firms were identical; that there were constant returns to scale in production (adding the same number of extra units of input increased output by a constant amount no matter what the existing level of output); and that there was perfect competition such that no firm could charge a price higher than its costs and make a level of profits higher than the minimum needed to stay in business. The reason for these assumptions had been for the sake of the tractability of the models. But a brilliant technical innovation introduced by Avinash Dixit and Joseph Stiglitz in a 1976 paper spread like wildfire throughout economics (Dixit and Stiglitz 1976). They found a very tractable way of introducing “monopolistic competition.” Firms each produce a differentiated product, and thereby have a bit of monopoly power thanks to this bit of distinctiveness. There are also increasing rather than constant returns to scale, which is more realistic as firms in most businesses have some fixed costs which can in some cases be very large. Fixed costs mean that there are economies of scale—the average cost of each item produced will be lower the larger the production run. In some industries, from aerospace to software, fixed costs are large. Paul Krugman took the next step, applying this kind of model to international trade, and subsequently to questions of industrial location in general (Krugman 1980, 1991a,b; Helpman and Krugman 1985). Economists had long realized that increasing returns were an important part of the explanation for why industries tended to locate in particular geographic clusters, or why people tended to congregate in towns and cities, which is where economic growth occurs. Urbanization is intimately linked to growth, and if the account of growth theory which follows ignores the economic geography literature, this is only for reasons of space (see Fujita et al. (1999), which also references this literature, and Glaeser et al. (1995)).

Common to the endogenous growth models of both Lucas and Romer is the introduction of *spillovers*, whereby individual decisions have an immediate effect on others. This is what provides an escape from the realm of diminishing returns, and therefore from the two empirically refuted neoclassical implications of convergence between countries and growth rates which decline over time. Indeed, with increasing returns, rather than diminishing returns, growth can snowball in a virtuous circle,

and economies can easily diverge. I'll look at spillovers in human capital and innovation in turn.

My earlier description of the neoclassical model referred to inputs of "labor" and mentioned, fleetingly, issues of labor quality, such as educational attainment. The right kind of concept to use in a theory of growth is not the number of workers but rather a quantity which also takes account of their skills and effectiveness. The term used is *human capital*, the accumulated stock of human abilities, by analogy with the accumulated stock of equipment that makes up physical capital. Lucas adapted the standard model to incorporate human capital explicitly, including individual decisions about how much to invest to acquire knowledge. He argued that such decisions are made in a social context, in a way which has no parallel for decisions to invest in machinery. If I live in a country where average levels of education are low, it is unlikely to pay me to invest heavily in my own human capital because the returns will be low: they will be geared toward the average, which my individual decision cannot affect. So countries can become settled in a low-level trap. Conversely, the payoff to my education will be greater if I'm working with other people who are highly educated. Here is a mechanism which can clearly lead to divergence, not convergence, between economies. In terms of the formal model, Lucas introduced a production function depending on physical capital, aggregate human capital, and average human capital, with diminishing marginal returns to the first two terms, and a spillover from average human capital to the other terms. The incentive for individuals to invest in their own human capital depends on the decisions of others.

It was the introduction of human capital into the theory by Robert Lucas which prompted Robert Barro to compare the growth performance of economies depending on their initial attributes, finding the conditional convergence described above. Starting poor means faster growth, and therefore catch-up to the richer economies as the neoclassical model of accumulation implies, as long as a measure of human capital is included. This result, widely confirmed by other researchers, makes the human capital approach look promising empirically. And of course it makes intuitive sense that improving standards of education are related to better economic outcomes. Barro pointed out that the poor countries which have decisively converged on the rich ones—East Asian "tigers" such as South Korea and Taiwan—had started out with much higher levels of human capital than countries with similar levels of GDP per capita in 1960. So endogenizing the accumulation of human capital in this way

is a big step forward. It is likely to be vitally important in thinking about growth in developing countries where initial levels of human capital are low. This is an important theme in the next chapter.

Yet education cannot have been decisive during the Industrial Revolution, when literacy levels were low, and many innovators hadn't been to school at all. So other economists have endogenized technology and the process of innovation instead. In another landmark paper Paul Romer (1986) introduced the economy's stock of knowledge into the production function, alongside the stocks of physical and human capital. In turn there is a separate production function for knowledge, which depends on the inputs each business devotes to accumulating knowledge and the externality—the economy's total public stock of knowledge. Each business has an incentive to invest in its own knowledge but part of that unintentionally becomes public, not just private, knowledge. New discoveries are to some extent a public good: once somebody makes the conceptual breakthrough, others will imitate. There are diminishing returns to capital and labor but not to aggregate knowledge. Indeed, there are *increasing* returns to knowledge: the same bit of knowledge can be used repeatedly without being used up; and one bit becomes the foundation for the next (no description has been more expressive than Isaac Newton's "standing on the shoulders of giants"). This formalization of growth clearly corresponds to the emphasis some economic historians such as David Landes and Joel Mokyr have placed on ideas and technology.

Romer extended his original model to incorporate research and development (R&D) spending and patenting by businesses. The temporary monopoly offered by a patent gives businesses the chance to profit from product innovation, but the act of patenting in itself puts part of their innovation into the public domain. He wrote: "Long-run growth is driven primarily by the accumulation of knowledge." The beauty of his model and its successors is that they provide a theoretical version of the lesson of history, that growth must be crucially linked with innovation, which can be tested empirically (see Grossman and Helpman 1991; Aghion and Howitt 1998). The empirical evidence certainly supports the importance of such R&D "spillovers" in explaining growth. Indeed, the links between innovation and growth have been explored in detail and there is now a huge literature documenting the existence and size of R&D spillovers in many contexts.¹⁰

¹⁰For a rich account of the process, see Baumol (forthcoming).

Stepping back from such detail, though, how well do the different endogenous growth models mesh with the very broad contours of economic growth? Do they overcome the empirical shortfalls of the Solow model? And what are we to make of the fact that there are different versions? One point to note before assessing the implications of endogenous growth is that the two models are essentially similar. The stock of knowledge in an economy will be at least partly attached to humans, as part of their human capital; and the parts of human capital which spill over into society, know-how that is "in the air," will be part of the public stock of knowledge. There is surely more progress to be made on understanding these links and measuring human capital and knowledge. But a rounded theory of growth must have room for both human capital and product innovation. Both are empirically important in explaining the growth differences between countries.

THE IMPLICATIONS OF MODERN GROWTH THEORY

Recall that two aspects of reality motivated the introduction of increasing returns to the two types of investment, in human capital and in knowledge. One was the evidence that growth in the rich countries has tended to accelerate over time. The other is that countries' growth records have diverged rather than converged. The basic Solow model, with diminishing returns to the inputs of labor and capital, is inconsistent with these two big facts, whereas endogenous growth models with increasing returns to scale are consistent with them. What are the implications of the presence of increasing returns in the growth process? A world of increasing returns is vastly different from Solow's "land of the margin" with diminishing returns. This step turns out to make size, history, and geography important in theory for economic outcomes: modern growth theory is contingent on circumstances in a way far removed from the abstract world of neoclassical growth.

Size

The use of knowledge is nonrival—ideas don't get used up but can be used repeatedly by many people. Thomas Jefferson famously compared the process to lighting one candle from another. And the more people there are, the more potential inventors there are, the greater the potential contact with new ideas, and the greater the scope for intellectual specialization. In addition, the per capita resource cost of creating knowledge

will be lower the larger the population. So the more people there are to use the available knowledge, the greater the economy's output can be. The more people there are, the more knowledge will be added to the public stock. Population growth in itself will spur technical progress and growth in *per capita* output. At least, that's the theoretical implication. It is controversial.

Eyeballing the very-long-term trends in global population and per capita GDP does suggest a link, as population growth started to accelerate around the same time as output. World population climbed slowly from just over a quarter of a billion in the year 1000 to about half a billion in 1600 and just over one billion in 1820, while it has jumped to six billion in the less than two centuries since. In a 1960 paper, three electrical engineers tested the proposition that population growth is proportional to the population level and found that it fitted the historical evidence well. Extrapolating forward, they predicted the world's population would become infinite on Friday November 13, 2026 (presumably an electrical engineers' joke, very much on a par with the sense of humor of a typical economist). Extrapolation is clearly inappropriate—a richer model is needed to predict future population growth.¹¹ Indeed, in the decades since 1960 the pace at which the number of people in the world is climbing has slowed to be much less than proportionate to the population level. Economist Michael Kremer tested, using a very long run of historical evidence, a model in which more people make for more innovation and faster growth, but in which population growth is also affected by income per capita. In other words, it incorporates the well-known demographic transition of a steep drop in fertility rates in countries which climb above a low level of income (Kremer 1993).

On the other hand, it is not true to say that big economies always grow faster than small ones; there are plenty of small but rapidly growing economies, and small but rich ones. Countries such as Singapore or Switzerland are examples. This fact has made economists a bit uneasy about the first wave of endogenous growth models, with their sizist implications, leading some to develop modifications to limit or eliminate the importance of scale. But, after all, the scale effects were introduced to make the model fit better with the evidence. If human capital is emphasized, the more people there are, the greater the knowledge spillovers

¹¹ One of the important differences between the approach typically taken by economists and those of other disciplines is that economists are far less likely to extrapolate a trend indefinitely forward; economic models inherently include adjustment mechanisms.

in the accumulation of human capital. If innovation is emphasized, the larger the market, the greater the profitability of investing in R&D, and the greater the spillovers. Either way, there is a chain reaction, a virtuous circle which starts slowly then accelerates, related to the scale of the economy. I don't want to overemphasize the importance of scale at all, but it's a wrinkle in the theory which makes some economists very uncomfortable.

Where endogenous growth models have clearly enriched our analysis is in incorporating the real-world complexity of place and time. It matters where an economy is located, and what its history has been. The accumulation of specific experience, reflected in an economy's institutions and norms of behavior, can have extremely long-lasting effects on growth performance. In part 3 of the book I return to these "microfoundations" of macroeconomic performance from the bottom-up perspective of individual choices and interactions. Here I consider their top-down role in growth theory.

Geography

The existence of human capital spillovers, whereby your skills and experience affect my productivity and earning capacity, has other strong implications. If people are free to move, those with human capital that's above average for their location will try to move somewhere with a higher average level. This might account for the enormous pressures for migration from poor to rich countries. It would certainly help explain the tight correlation between economic growth and urbanization. For why do people crowd together in cities, with their filth, congestion, danger, and of course high prices? Costs of production are higher in cities than in less densely populated areas. The only plausible reason for urban concentration is to be close to other people. We learn from others, we exchange ideas, they stimulate us to achieve more ourselves. Alfred Marshall made this argument in a well-known passage, in his 1890 textbook, about the clustering of economic activity in industrial districts:

The mysteries of the trade become no mysteries but are as it were in the air, and children learn many of them unconsciously. Good work is rightly appreciated, inventions and improvements in machinery, in processes and the general organization of business have their merits promptly discussed. If one man starts a new idea, it is taken up

by others and combined with suggestions of their own; and thus it becomes a source of further new ideas.¹²

Marshall (1920)

The great urbanist Jane Jacobs has long emphasized the economic role of cities, and the importance of the creation of public knowledge in industries such as finance, clothing and textiles, autos, or advertising (Jacobs 1985). And geographers have always, naturally, been interested in questions of location. The introduction of increasing returns in growth theory has generated new insights into the economic analysis of geography, whether that's regional disparities or patterns of trade between countries. As well as addressing the question of how economies grow over time, economists can say more about how they extend in space: for example, why does one city thrive and another decline? The existence of specialized industrial clusters such as shipbuilding or computer software is well-known, but why do they form and why are they where they are? A natural harbor might explain a shipping industry—but a programmer could hunch over a screen anywhere with electricity and telecommunications.

Paul Krugman, although now a famously polemical *New York Times* columnist, made his name applying increasing-returns models to such locational questions, in one of the first examples of how these models came to characterize a new way of thinking about the economy in general (see Krugman (1980), with further extensions and references in Krugman (1991a)). I first met him when I was working as an economics journalist and wanted to interview him about the exciting new developments in economic geography; but setting up the interview was hampered by the fact that, although he knew he was in London, he couldn't tell me his own location. As he'd recently come from giving a lecture in Tierra del Fuego on the far southern tip of Argentina, finding him in my own city seemed feasible, and indeed I tracked him down. Krugman's insight, which sowed the seeds of an intellectual revival in the entire field of economic geography, was that just as increasing returns explain divergent growth histories, they account for divergent geographies as well. The

¹²Marshall of course had the sensibilities of his time, but this usage does remind me of one of my favorite cartoons (from *Punch* magazine), of a board meeting with a sole woman present. The chairman says: "That's an excellent suggestion Miss Triggs. Perhaps one of the men would like to make it."

virtuous and vicious circles of growth occur in particular places, in one country (or region or city) rather than another.¹³

To spell it out, knowledge spillovers are made possible by the physical proximity of people, especially when it comes to the sharing and exchange of complicated and abstract ideas, whether in software or a service such as finance or medicine. This is why business executives still fly thousands of miles for face-to-face meetings, why universities exist, and why computer experts cluster together in Silicon Valley. Human capital spillovers (positive or negative) are similarly amplified by the near presence of others. The return to my education is more likely to depend on the educational attainments of other people if we are all present in the same job market. I will only have the prospect of employment in a well-paid job if there is a pool of skilled labor which will attract the kind of employers who want my kind of skills. Places without the skilled people will fail to attract such businesses; and without those good job prospects, few people will have the incentive to ever acquire the skills the high-value firms need. The catch-22 is easy to understand, but very hard to escape. Call it the development catch-22—it sheds new light on the failure of the poor countries to grow, as we will see in the next chapter.

It might seem banal, in that uniquely obscure academic way, to say that economic growth occurs in specific places. However, the analysis of growth up to the end of the 1980s was abstract in the extreme. So one of the spillover benefits of endogenous growth theory, as it were, has been to restore to its proper place the importance of detail. Within a very general analytical framework, you can nevertheless expect a different growth performance from different places.

History

It is not only the accidents of geography that matter, but the accidents of history too. Once a pattern of growth or stagnation becomes established in an economy, whether of an individual town or a whole nation, it takes a lot to make it change. Actually, there is an asymmetry: it's easier to stop a snowball than to start one. Big events such as wars or radical innovations can derail prosperous economies even where the geographical factor is preeminent: for example, the combination of the Civil War

¹³ His model also included transportation costs, bizarrely ignored in earlier trade theory. *Geography and Trade* (1991) is a very accessible introduction. *The Spatial Economy* (Fujita et al. 1999) is a key textbook. In the context of urban growth and decline, see Glaeser (2003). The business analysis of locational clusters is due to Porter (1990) and Kay (1993).

blockades and the invention of kerosene in place of whale oil destroyed the New England whaling industry. (The final straw came when women dropped the whalebone corset into the dustbin of history (see Menand (2002) for these details).) Setting a snowball rolling, on the other hand, involves the coordination of very many individual decisions to offset adverse spillovers or reinforce favorable ones.

The importance of historical accident in the growth record has come to be widely recognized, and once alerted to this phenomenon, you see it everywhere. In the technical literature, it's called *path dependence*: where you are now depends on how you got there. One very-well-known example is the persistence of the QWERTY layout for keyboards, no longer needed on a modern computer keyboard as opposed to an old-fashioned typewriter, but impossible to change because anybody who has learnt to type is habituated to it.¹⁴ Another is the fact that the early luminaries of the modern computer industry attended Stanford University rather than, say, Chicago or Toulouse. Examples of historical happenstance are widespread in the business economics literature. Almost any industry you care to think of develops in geographical clusters, from the financial center of the City of London in the eighteenth century to ceramic tiles in northern Italy in the twentieth century, to call centers and business-process outsourcing in twenty-first-century Bangalore.

None of this is to say that either history or geography is destiny. How an economy got to where it is today (in both the literal and metaphorical senses) will certainly limit the range of options for tomorrow. But within that range many outcomes may be possible. By definition, if there are increasing returns to some activity such as acquiring human capital, then the more of that activity people undertake, the higher the return they get. There may be many possible paths of growth depending on their choices, or *multiple equilibria*. But what determines the actual collective outcome? One answer may well be the accidents of history, and this is the explanation which has predominated in recent work on growth. On the other hand, sufficiently strong expectations of a different turn of events could prove self-fulfilling: people might be persuaded for any number of reasons that their returns to investing in human capital or R&D in future will be higher than they have been in the past. A sufficiently determined education or industrial policy, for example, might

¹⁴ Anybody who doubts the power of this explanation should try typing on the keyboard of a French computer to learn how frustrating it is when just four characters are in a different place. See David (1985), Arthur (1989), and Arthur et al. (1997) on the importance of history for growth.

shift expectations toward such a self-fulfilling growth prophecy. This analysis explains why policy makers have grown increasingly interested in setting out formal growth strategies: they want to persuade their citizens to buy in to a collective vision of a brighter future because if they don't believe it, it won't happen.¹⁵

Disneyland economics? Well, although self-fulfilling prophecy is theoretically possible, policy makers can't just wish upon a star to get the growth rate they desire. Paul Krugman used a formal endogenous growth model to work out when hopes for the future might outweigh the legacy of the past, and showed that there are three conditions (Krugman 1991c). First, there must be enough interdependence between individual decisions to start with. Second, people must put a relatively high weight on the future compared with the present (there must be a sufficiently low real discount rate), because obviously if they discount the future too much it will not have much influence on their decisions today. Finally, the economy must be able to adjust reasonably quickly, because if it adjusts slowly, the shadow cast by history will always dominate. To put it another way, people must want to change, their wishes must have some traction on the economy, and must be mutually reinforcing. In any particular economy, it will be possible to be more precise about the magnitudes involved. But in any case, these conditions all depend on deep-rooted social and behavioral characteristics, which rules out quick policy fixes.

Institutions

An appreciation that growth is a collective outcome, depending on the interdependence between individuals' choices, in fact makes institutions absolutely central. After all, an institution is by definition a means for people to combine together. The importance of institutions for economic development will be a theme of the next chapter, and I will also revisit it in detail in chapter 8. An economic historian who emphasized the importance of institutions long before it became fashionable to do so in development circles was Douglass North, cowinner of the 1993 Nobel memorial prize (with Robert Fogel). In a much-cited 1968 paper, he showed that productivity improvements in the shipping industry depended more on organizational than technical change (see Fogel (1999) for a review). There are echoes of his historical result in recent

¹⁵ See Bradley (2005), for example, for an account of how the Irish government put this into practice in the 1980s, turning the country into the "Celtic tiger."

research in U.S. productivity growth in the 1990s, which also emphasizes the preeminence of organizations over technological innovations alone (see, for example, Brynjolfsson and Hitt (2000) and McKinsey's Global Institute (2001) and the extensive references contained in them). In a nutshell, institutions matter because they are the vehicles for transmitting the legacy of history.

In a brilliant paper Mancur Olson noted that due to their institutional weaknesses, poor countries have extremely inefficient economies. They do not use their available physical and human capital or available technology very well. An immigrant from Haiti to the United States becomes instantly more productive when she lands, indicating that her human capital was not being fully utilized at home. Returns to investment in physical capital in developing economies are very high, but the investment doesn't take place. The plausible reason is that social institutions in poor countries inhibit growth. This fits in with the fact that a few poor countries do grow extremely rapidly for some period: they have a catch-up spurt which takes them to the efficient frontier which the rich countries occupy. At this frontier, endogenous growth models do a very good job of fitting with the facts (Olson 1996).

The fact that the process of growth is not the same everywhere and always, that in fact the outcome is highly contingent, complicates giving policy advice. It is hard to make any generalizations beyond the obvious point that growth depends on the collective capacity of a society, through its institutions and its current endowment of human and other resources, to accumulate and combine resources. Indeed, economists still have some serious disagreements about how high the long-term growth rate is in the world's most intensively measured and scrutinized economy, the United States. Table 1.1 indicated that long-term growth has tended to accelerate over time. Many, but not all, economists would argue that there has been a further acceleration since the mid 1990s, thanks to the innovation of the microprocessor.

BACK TO THE FUTURE

Given that how economists think about economies has depended on the context of their time, it should not be surprising that growth theory saw a striking revival in the late 1980s. After all, it was then that many people, especially academics, started to notice the arrival of computers on their desktops and wonder whether the massive increase in computer power and declines in its price might have an impact on the economy. A decade

later, nobody, surely, could have failed to notice the "New Economy," an ill-defined phenomenon which nevertheless required everybody to define themselves as either a skeptic or an enthusiast.

New technologies have always given rise to some apprehensions. Mary Shelley played on fears of the power of electricity in *Frankenstein*. It is quite amusing now to come across the description of "The mighty agency of steam, capable of almost unlimited good or evil," in Elizabeth Gaskell's *Mary Barton*, but such views were a Victorian commonplace. While few people seem to fear computers or mobile phones in quite the same way, there has certainly been a clear tendency to belittle their impact.¹⁶ In 1986, Robert Solow (he of the workhorse neoclassical growth model) famously introduced the "productivity paradox": this new technology appearing in workplaces across America seemed to have no impact on productivity and therefore long-term growth. It was not until the late 1990s that measured growth seemed to have increased at all, and it took another few years for a majority of economists to accept that the long-term average for U.S. productivity growth had increased. Indeed, as the evidence that there was in fact a productivity impact from the technologies started to emerge, skepticism took new forms. Robert Gordon, for example, argued that computers alone could never have the impact of past waves of innovation, which had involved more than one new technology emerging at the same time. His example was the combination of innovations in 1860-1900 leading to electricity, motor and air transport, radio and movies, and indoor plumbing (Gordon 1999, 2000).

There are several responses to the argument that computers have not been very important for growth. One is that measuring the impact of steam or electricity in exactly the same way as the impact of computers is measured (using the growth accounting described above), you find that steam and electricity look pretty small too: a "small" percentage point difference in growth rates is the statistical footprint of a large economic and social change (Crafts 2004). (Recall from the last chapter that the Industrial Revolution doesn't look very revolutionary in the GDP growth statistics.) A second is that any radical innovation takes a long time to have a measurable aggregate impact because people take many years to adjust: perhaps new infrastructure must be built, new skills learned, workplaces reorganized (David 1991). Indeed, many people have an incentive to resist innovations. As Niccolò Machiavelli put

¹⁶ And some people obviously do have serious apprehensions about biotechnology and genomic research.

it in *The Prince*, "Innovation makes enemies of all those who prospered under the old regime, and only lukewarm support is forthcoming from those who would prosper under the new." And, lastly, although popular attention has focused on computers, there is a cluster of new technologies today, including biotechnology, new materials, and nanotechnology. Their combined impact on our well-being is likely to be just as profound as the cluster of technologies introduced around the start of the twentieth century.

I've always been puzzled that some economists have been so skeptical that huge declines in the cost of processing and communicating information will affect the allocation of resources and organization of the economy, when information is fundamental to economic analysis (see chapter 6). Still, this is not the place to tramp further over this already well-trodden ground.¹⁷ Suffice it to say that there is now a consensus that computer-related technologies have led to an acceleration in long-term growth in the United States and a few other economies since the mid 1990s. But there is little agreement about why most countries, especially the other leading Western economies, have not yet shared in this phenomenon. The reasons for the difference must lie in the kind of behavior and institutions which prevail in the United States rather than in Japan or Germany, and attention has focused on factors such as the ease of financing a new venture, patent rules, the science base in universities, and the availability of people with suitable skills. Will Baumol (2002) has used the marvelous image of an "innovation machine" to capture the idea of a whole economic system whose essence is inventing and implementing new technologies. He says of the "unparalleled growth performance of the capitalist economies":

The free market, once the institutional impediments to its development had been reduced sufficiently, just grew by itself, and by itself became the machine that generates innovation and growth in dramatic profusion.... It is clear that innovation plays a far larger role in the activities of many key firms and industries than the current theoretical literature takes into account.

Baumol (2002, pp. 2, 3)

This emphasis on the importance of technological innovation is an overdue return to the lessons of the Industrial Revolution. Increasing-

¹⁷ I have already done so myself in two books, *The Weightless World* (1997) and *Paradoxes of Prosperity* (2001). The best recent summary overview is in Helpman (2004) and the OECD has several publications making an empirical assessment of the links between new technologies and productivity growth in member countries.

returns models of endogenous growth theory, along with the emphasis on innovation, now inform an extensive research agenda in many fields of economics. Thanks to the theoretical and empirical advances of the past twenty years, economists now have a good understanding of the process of growth. Although endogenous growth models are even more mathematically rigorous and formal than their predecessor neoclassical version, and for all the jokes about the name, modern analysis of economic growth is profoundly less abstract than it used to be. The realm of endogenous growth theory is a universe away from the neoclassical competitive equilibrium model, as it puts the specifics of time and place, and the gritty details of particular economies, at center stage. As a result, the new theories have completely changed the focus of economic policy makers when it comes to asking how to boost productivity and growth, with the emphasis now on education, research, innovation, the structural flexibility of the economy, and the quality of economic institutions.

This chapter has looked at some quite general theories of long-run economic growth. These models have their roots in similar models which have transformed just as thoroughly the way economists analyze trade and industrial structure. These concern the fundamental mechanics of the economy and will ultimately be integrated with growth theory, which at present concentrates on ideas and the mechanisms of knowledge transfer. Recall from chapter 1 that there are two keys to modern growth, according to the historical evidence: technical innovation which improves productivity in certain industries, and increasing specialization through trade. How firms in the industries where an innovation becomes available decide what to make and how to price their products and how it comes about that firms in one country or another reap the fruits of their specialization through international trade are vital pieces of the growth jigsaw. In a spectacularly fruitful period of mutual exchange of ideas, the economists working on endogenous growth models and those working on new models in industrial organization and trade transformed all three areas of the subject in the mid to late 1980s. The typical textbook model of an economist now is a world away from the simple textbook models taught a generation ago. No longer does economics assume identical firms and perfect competition. Firms are typically assumed to be "monopolistically competitive," with a degree of monopoly power gained through differentiation of their products (signaled perhaps by trademarks, or brands) in order to cover their fixed costs of production; but open to competition because entry into the

industry by new firms will occur if monopoly profits rise too much. As well as variety, there's innovation. Will Baumol (forthcoming) has integrated the theory of entrepreneurship into these formal models of production, making innovation fundamental to how firms compete with each other. His new textbook based on this modeling approach will take its place as the standard for training the next generations of economists, just as his *Economic Theory and Operations Analysis* set out the state of knowledge on price and production theory for my generation when it was published (Baumol 1977). The revolution in the very basics of microeconomic theory has hardly been appreciated at all outside the profession.

In particular—and thanks in part to our recent experience of the ways new information and communication technologies, biotechnology, and other technologies are affecting our own societies—economists have made huge strides recently in analyzing the process of innovation. Will Baumol has explored the differences between radical innovation by entrepreneurs and incremental improvement for the mass market by big corporations, and has brought the process of entrepreneurship within the ambit of standard microeconomic theory. Economists are exploring the process of patenting innovations, the best way to protect intellectual property in order to encourage innovators; the implications for productivity of inward investment by leading foreign companies and the knowledge spillovers which can result; the paths through which research in universities is put into commercial practice by entrepreneurs; and many other empirical details of how it is that technological change turns into aggregate growth.¹⁸ The microeconomic undergrowth beneath the canopy of the macroeconomic growth literature is steadily being filled in by researchers, with important implications for government policy. These processes are amenable to government intervention in ways that a country's history, culture, and geography are not. I'll return to the importance of modern microeconomic research later. But I want to round off this part of the book by applying the lessons of economic history and growth theory to the question of global poverty. What can economics say about why the poor economies haven't grown?

¹⁸ The empirical microeconomic literature on innovation and growth, and the behavior of inventors, entrepreneurs, and firms is vast. Literally dozens of new research papers (in English, never mind other languages) are published each week. As a starting point, and for further references, see Audretsch (2006), Jaffe and Trachtenberg (2002), and Jaffe and Lerner (2004) on patents and the papers by Baumol (2002, forthcoming) on innovation and entrepreneurship.